Università di Pisa

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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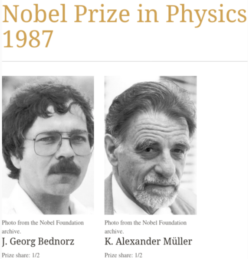
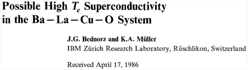
Outline

Cuprates and Hubbard physics




Possible High T_c Superconductivity in the Ba – La – Cu – O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

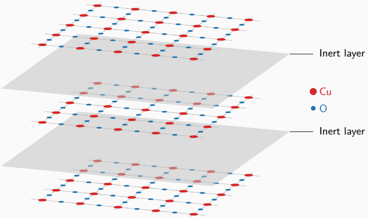
What makes cuprates interesting?

- Strong "unconventional" insulators;
- Rich phase diagram;
- Origin of Cooper pairing: still unclear;
- Highest critical temperatures for SC transition in this class of materials.

Cuprates and the Hubbard model

These materials are stackings of:

- 2D copper oxide active layers;
- intermediate layers of inert material.



Most of the cuprates physics is well described by the single-band Hubbard model on the square lattice:

$$\hat{H}_{\mathrm{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \qquad \text{with } t, U > 0.$$

The phase diagram of cuprates

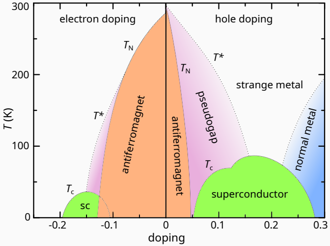
Cuprates exhibit various phases, among which:

- **Antiferromagnetic long-range order** at null and low doping;
- d-**wave superconductivity** as we inject or remove charge.

At **half-filling**, band theory would predict a metallic state. Instead we observe strong insulation.

$$\Downarrow$$

Cuprates are **not** band insulators.



Schematic phase diagram of cuprates. Author: Holger Motzkau, Wikimedia commons, CC BY-SA 3.0.

Mott localisation: a many-body insulator

Band theory assumption: e-e interactions are **negligible**.

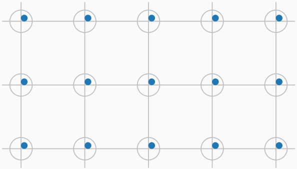
Strongly coupled Hubbard model: on-site repulsion

$$U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \,,$$

much bigger than kinetics. **Big energetic cost** of having two electrons on the same site.

$$\Downarrow$$

Insulation mechanism: electrons are **localised** as in a sort of “traffic jam”.



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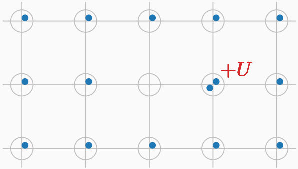
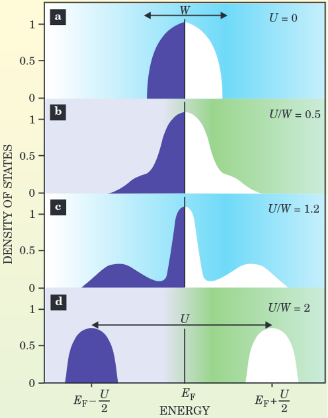


Bands flattening: a signature of Mott localisation



Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

(a) Free model, conventional DoS;

(b) Weak coupling, large coherent peak;

(c) Intermediate coupling, **narrow coherent peak**;

(d) Strong coupling, lower and upper Hubbard bands.

$$\Downarrow$$

Localisation = flattening of bare bands.

How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Perturbation theory at strong coupling gives:

$$\hat{H}_{\mathrm{HM}} \simeq J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E \,,$$

with antiferromagnetic exchange $J = 4t^2/U$.

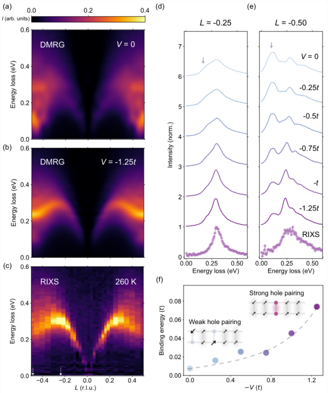
"Mottness": **the ground manifold is localised**.

Superconductivity

Phenomenology: we observe d-wave superconductivity, i.e. the superconducting gap is anisotropic under $\pi/2$ rotations $\mathcal{R}$,

$$\left\langle \hat{c}^{\dagger}_{\uparrow\mathbf{r}} \hat{c}^{\dagger}_{\downarrow\mathbf{r}+\Delta\mathbf{r}} \right\rangle = - \left\langle \hat{c}^{\dagger}_{\uparrow\mathcal{R}\mathbf{r}} \hat{c}^{\dagger}_{\downarrow\mathcal{R}(\mathbf{r}+\Delta\mathbf{r})} \right\rangle \,.$$

"Mottness": **non-local Cooper pairing**.



Hari Padma et al., 2025. **Physical Review X 15 (2): 021049**.

Incompatibilities with the spectrum of magnetic excitation in cuprates.

$$\Downarrow$$

Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\mathrm{EHM}} = \hat{H}_{\mathrm{HM}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V} .$$

We know from BCS theory that $\hat{H}_V$ can act **as a source of d-wave superconductivity**.

The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0.$$

The motivation: Recent findings of compatibility with this model for experimental results.

The idea: we know that the EHM can host d-wave superconductivity, which is related to Mott physics. Can we obtain manifest signatures of Mott localisation in the ground-state?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0.

Variational principle:

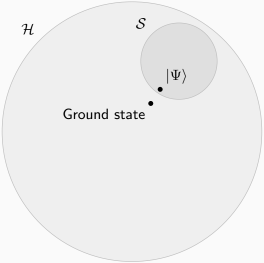
$$\forall\, |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} \quad : \quad E[|\Phi\rangle] \geq E_0\,.$$

$$\Downarrow$$

Constrained minimisation procedure:

$$|\Psi\rangle \quad : \quad \frac{\delta}{\delta\,|\Phi\rangle} E[|\Phi\rangle]\bigg|_{|\Phi\rangle = |\Psi\rangle} = 0\,.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

Consider all possible fermionic non-interacting hamiltonians:

$$\hat{K} \equiv \sum_{\alpha} E_{\alpha} \hat{\gamma}^{\dagger}_{\alpha} \hat{\gamma}_{\alpha} \quad \text{with} \quad \left\{ \hat{\gamma}_{\alpha}, \hat{\gamma}^{\dagger}_{\beta} \right\} = \delta_{\alpha\beta} \,,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

Hartree-Fock method: we identify $\mathcal{S}$ with the set of all the ground states of these hamiltonians,

$$\mathcal{S} = \left\{ |\Psi\rangle \;\middle|\; |\Psi\rangle \text{ is the ground-state of some } \hat{K} \right\} .$$

$$\Downarrow$$

Wick's theorem:

$$\langle \hat{c}^{\dagger}_{\alpha} \hat{c}^{\dagger}_{\beta} \hat{c}_{\gamma} \hat{c}_{\delta} \rangle = \underbrace{\langle \hat{c}^{\dagger}_{\alpha} \hat{c}_{\delta} \rangle \langle \hat{c}^{\dagger}_{\beta} \hat{c}_{\gamma} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^{\dagger}_{\alpha} \hat{c}_{\gamma} \rangle \langle \hat{c}^{\dagger}_{\beta} \hat{c}_{\delta} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^{\dagger}_{\alpha} \hat{c}^{\dagger}_{\beta} \rangle \langle \hat{c}_{\gamma} \hat{c}_{\delta} \rangle}_{\text{Cooper}} .$$

The self-consistency mapping

The iterative algorithm

That's all, thank you!

Backup slides